

DTU





Introduction to environmental indicators

Carbon footprint

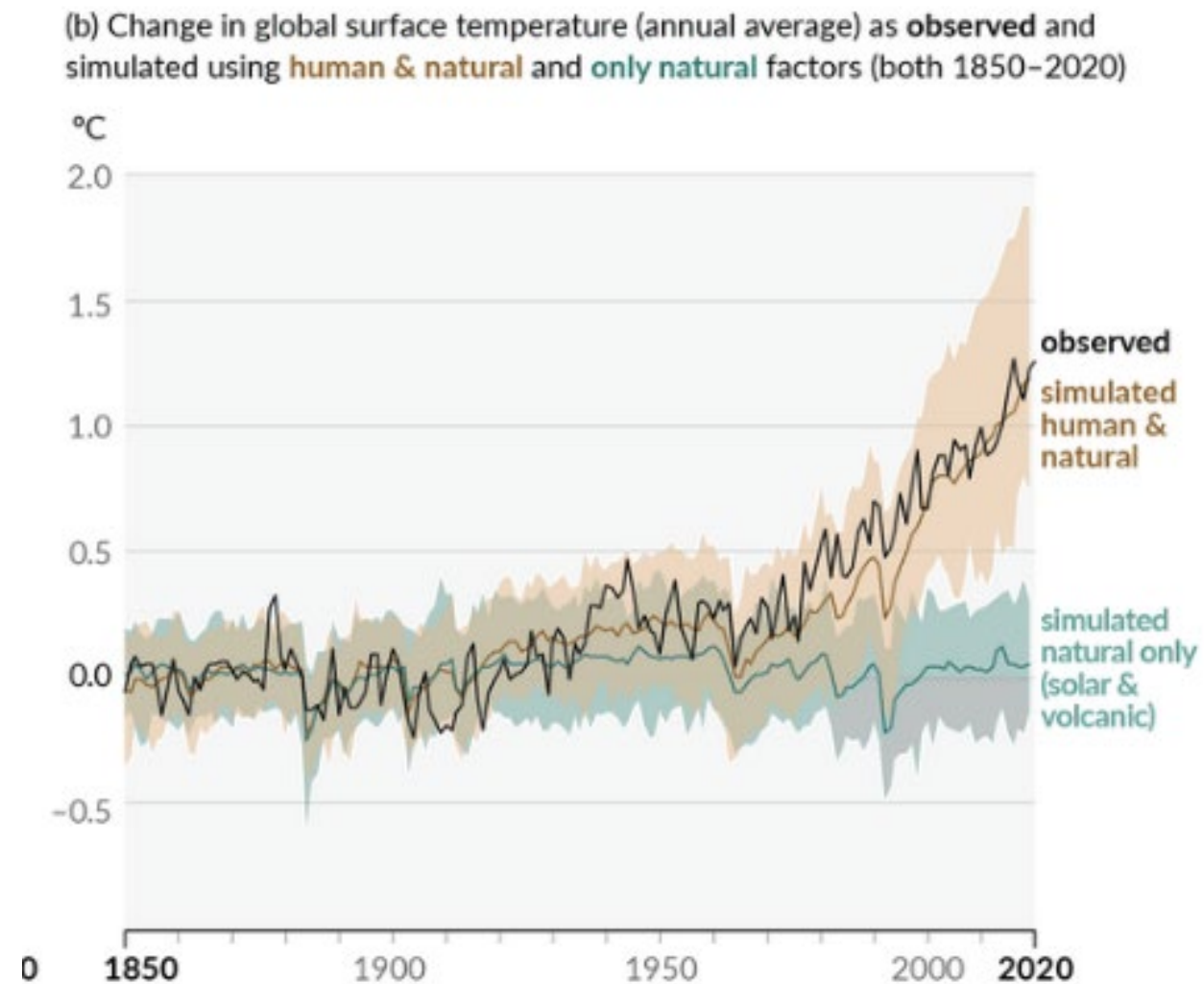
Olivier Jolliet, DTU-Sustain, Quantitative Sustainability Assessment

Learning objectives

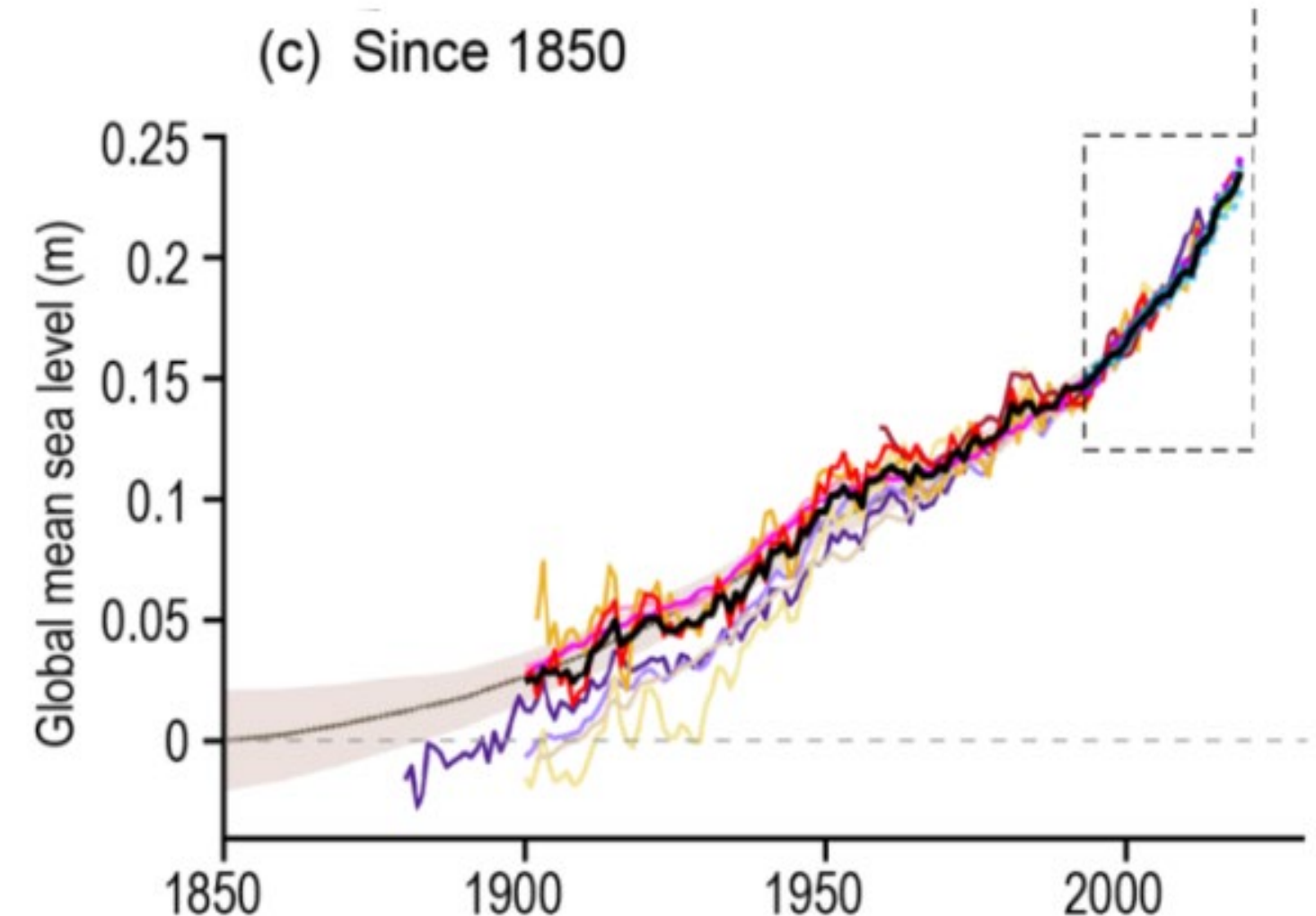
- ▶ Apply a causal framework to relate human activity to climate change indicators
- ▶ Calculate carbon footprint to determine how different substances can be aggregated based on impact
- ▶ Position the carbon footprint vs other indicators

Global warming – Diagnostic and causes

Global warming relative to 1850-1900
(°C)

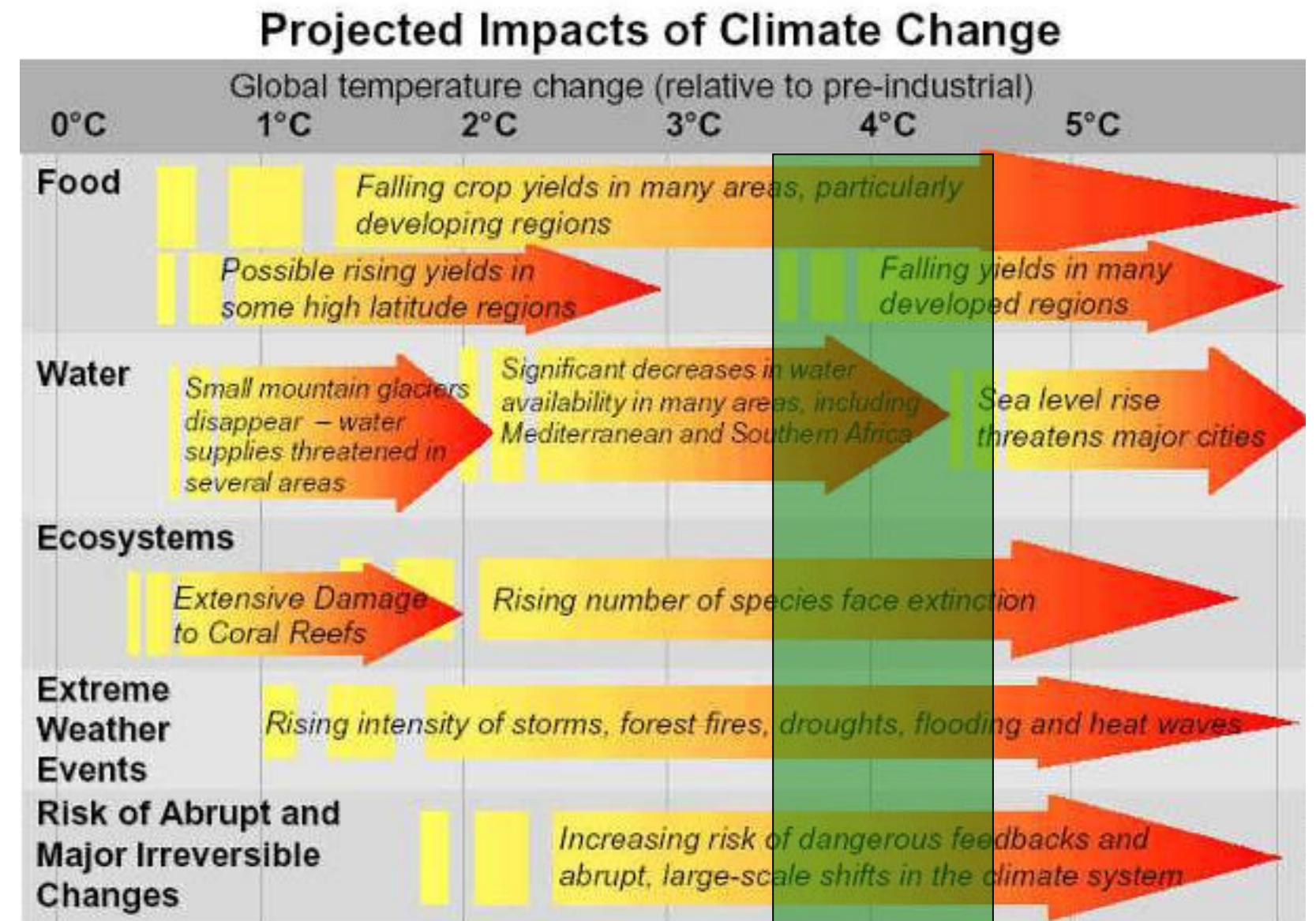
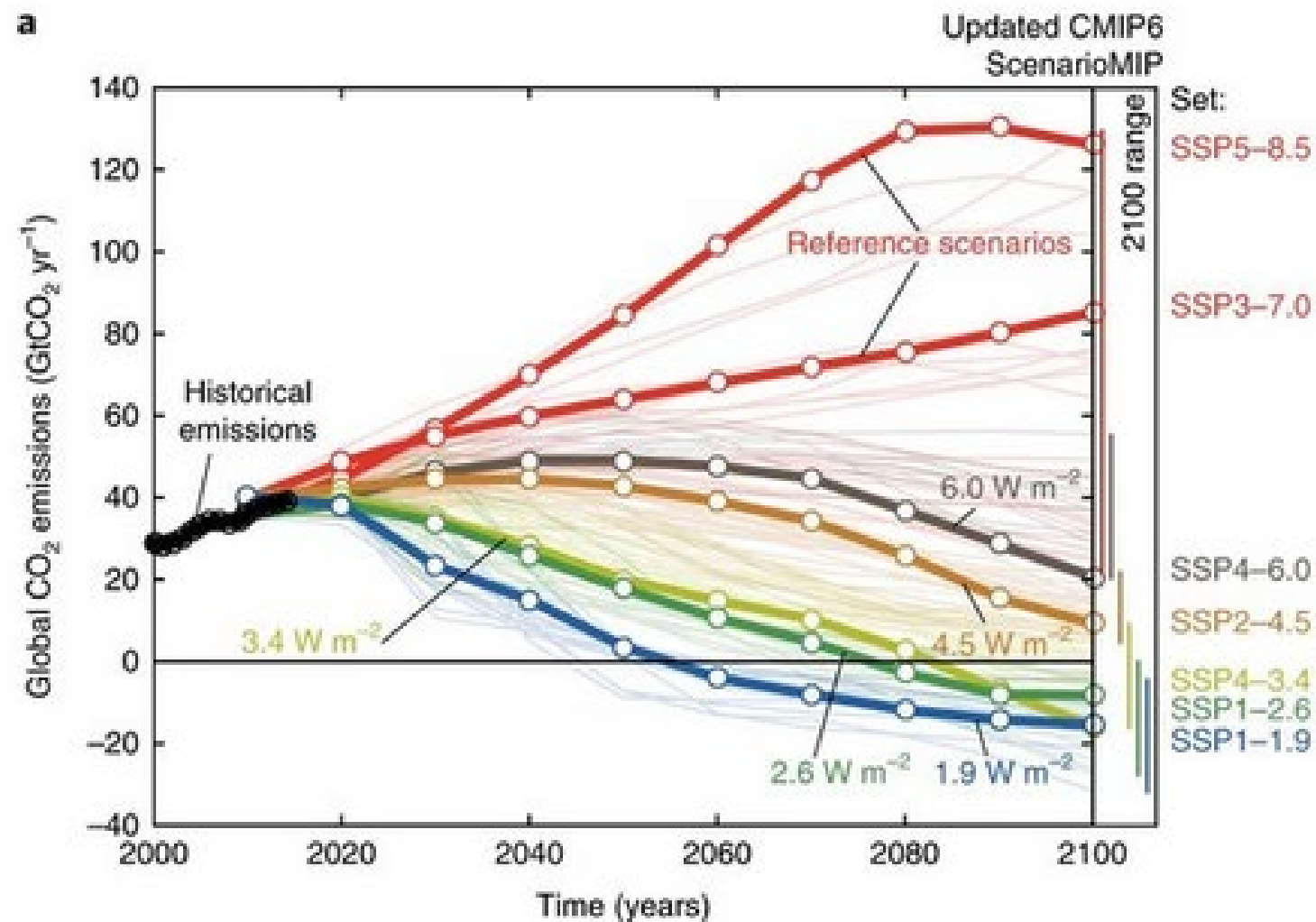


(c) Since 1850



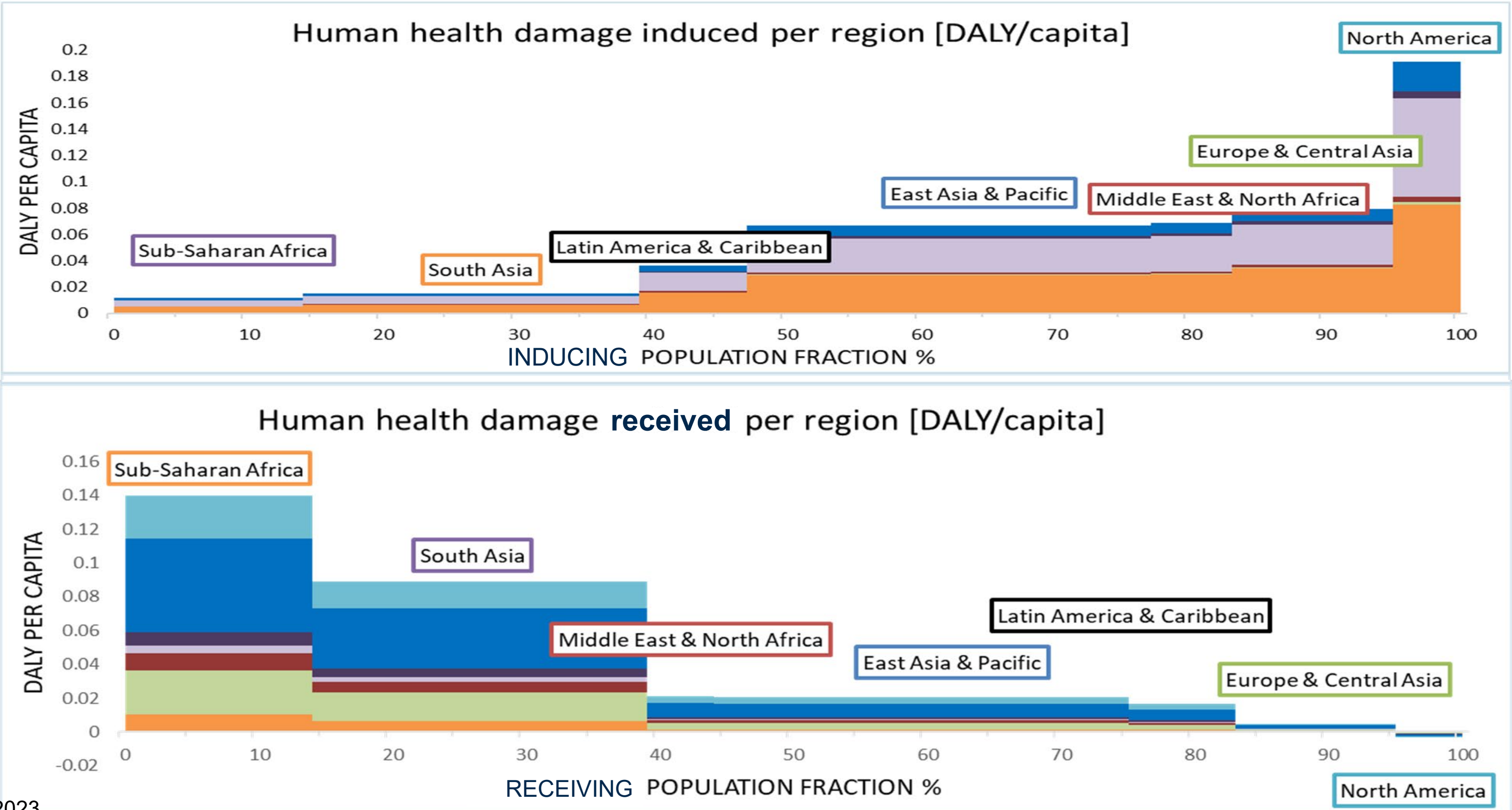
(IPCC, October 2019, SPECIAL
REPORT: GLOBAL WARMING OF 1.5
°C Summary for Policymakers")

The emissions projections and related impacts

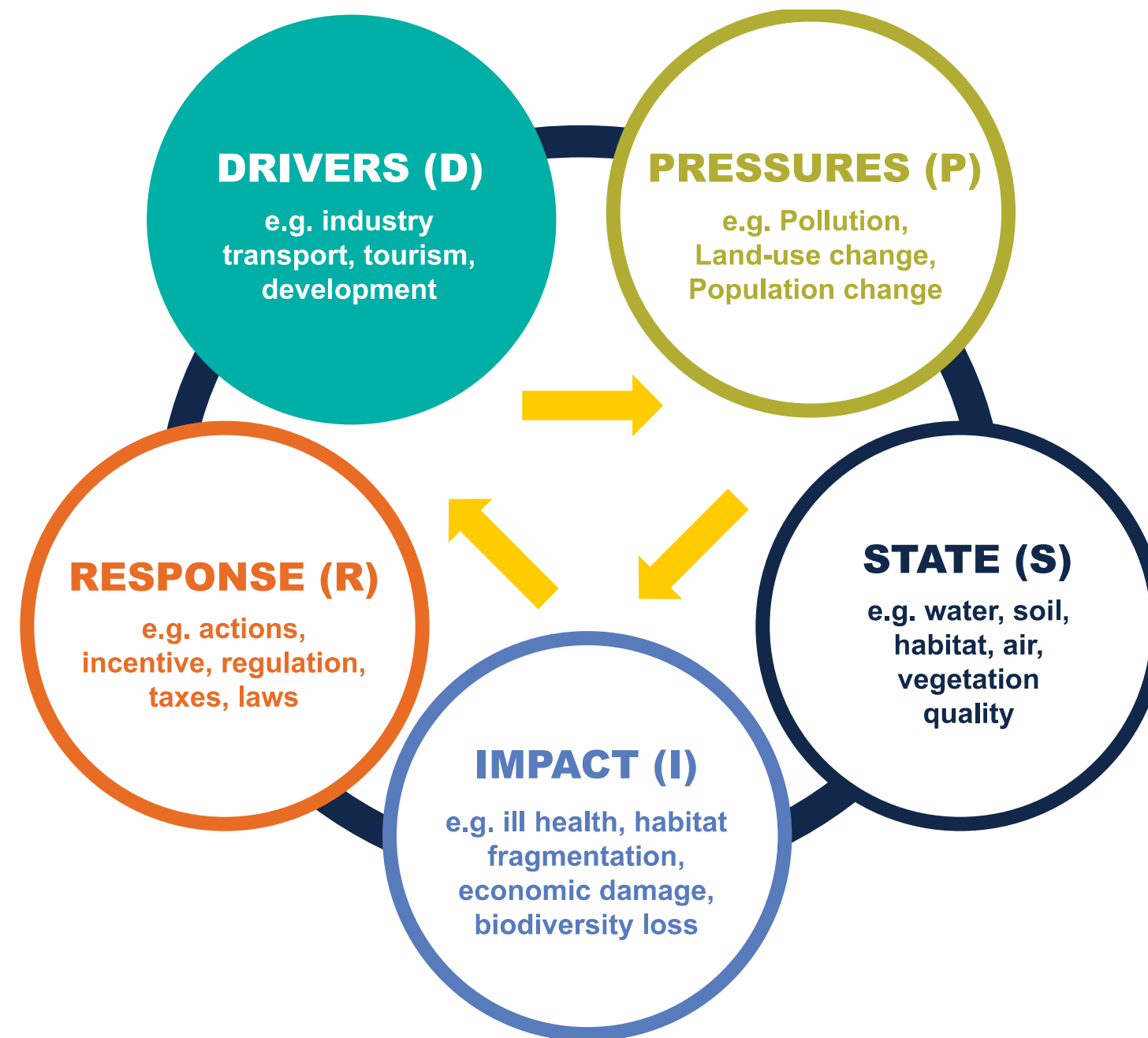


Likely range of 'business as usual' by 2100

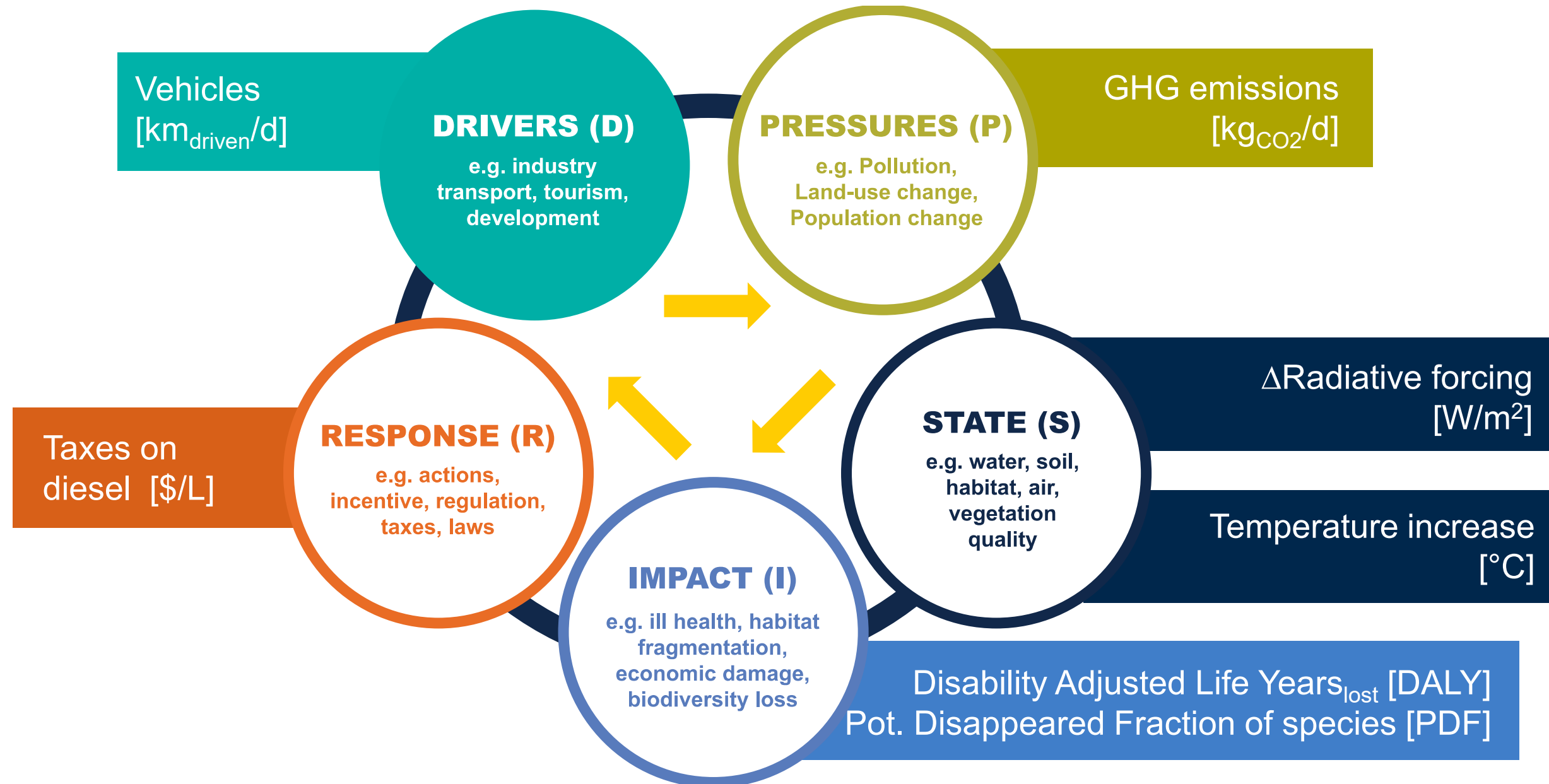
Disparities: Climate change impacts of heat and cold on health



The DPSIR model: to categorize and causally link indicators



The DPSIR model: applied to climate change impact of transportation



Carbon Footprint: at personal and product levels

A **carbon footprint** is the weighted sum of greenhouse gas (GHG) emissions caused by an individual, event, organization, service, or product, expressed as carbon dioxide equivalent. GHG emissions are weighted based on their global warming potentials (GWP100, as defined by the Intergovernmental Panel on Climate Change, AR6, Tables 7.15 and 7.SM.7)



Greenhouse Gas	GWP100
Carbon Dioxide (CO ₂)	1 kg _{CO2e} /kg _{CO2}
Methane biogenic (CH ₄)	27 kg _{CO2e} /kg _{CH4}
Methane fossil (CH ₄)	29.8 kg _{CO2e} /kg _{CH4}
Nitrous Oxide (N ₂ O)	273 kg _{CO2e} /kg _{N2O}
Hydrofluorocarbons (HFC-32)	771 kg _{CO2e} /kg _{HFC}
Perfluorocarbons (PFC-14)	7380 kg _{CO2e} /kg _{PFC}
Sulphur Hexafluoride (SF ₆)	24300 kg _{CO2e} /kg _{SF6}

Average DK carbon footprint per capita

Substance		Unit kg _i	Emissions kg _i /pers/y	GWP100 kg _{CO2e} /kg _i	GW score kg _{CO2e} /pers/y
Carbon dioxide	fossil	kg CO ₂	8000*	1	8000
Methane	biogenic	kg CH ₄	46.1		
Methane	fossil	kg CH ₄	4.7		
Nitrous Oxide*		kg N ₂ O	3.2		
HFCs, PFCs,SF6					80
TOTAL kg _{CO2e} /pers/y					
TOTAL kg _{CO2e} /pers/d					

*consumption based emissions

Average DK carbon footprint per capita

Substance		Unit kg _i	Emissions kg _i /pers/y	GWP100 kg _{CO2e} /kg _i	GW score kg _{CO2e} /pers/y
Carbon dioxide	fossil	kg CO ₂	8000*	1	8000
Methane	biogenic	kg CH ₄	46.1	27.0	
Methane	fossil	kg CH ₄	4.7	29.8	
Nitrous Oxide*		kg N ₂ O	3.2	273	
HFCs, PFCs,SF6					80
TOTAL kg _{CO2e} /pers/y					
TOTAL kg _{CO2e} /pers/d					

*consumption based emissions

Average DK carbon footprint per capita

Substance		Unit	Emissions	GWP100	GW score
		kg _i	kg _i /pers/y	kg _{CO2e} /kg _i	kg _{CO2e} /pers/y
Carbon dioxide	fossil	kg CO ₂	8000*	1	8000
Methane	biogenic	kg CH ₄	46.1	27.0	1250
Methane	fossil	kg CH ₄	4.7	29.8	140
Nitrous Oxide*		kg N ₂ O	3.2	273	870
HFCs, PFCs,SF6					80
TOTAL kg _{CO2e} /pers/y					
TOTAL kg _{CO2e} /pers/d					

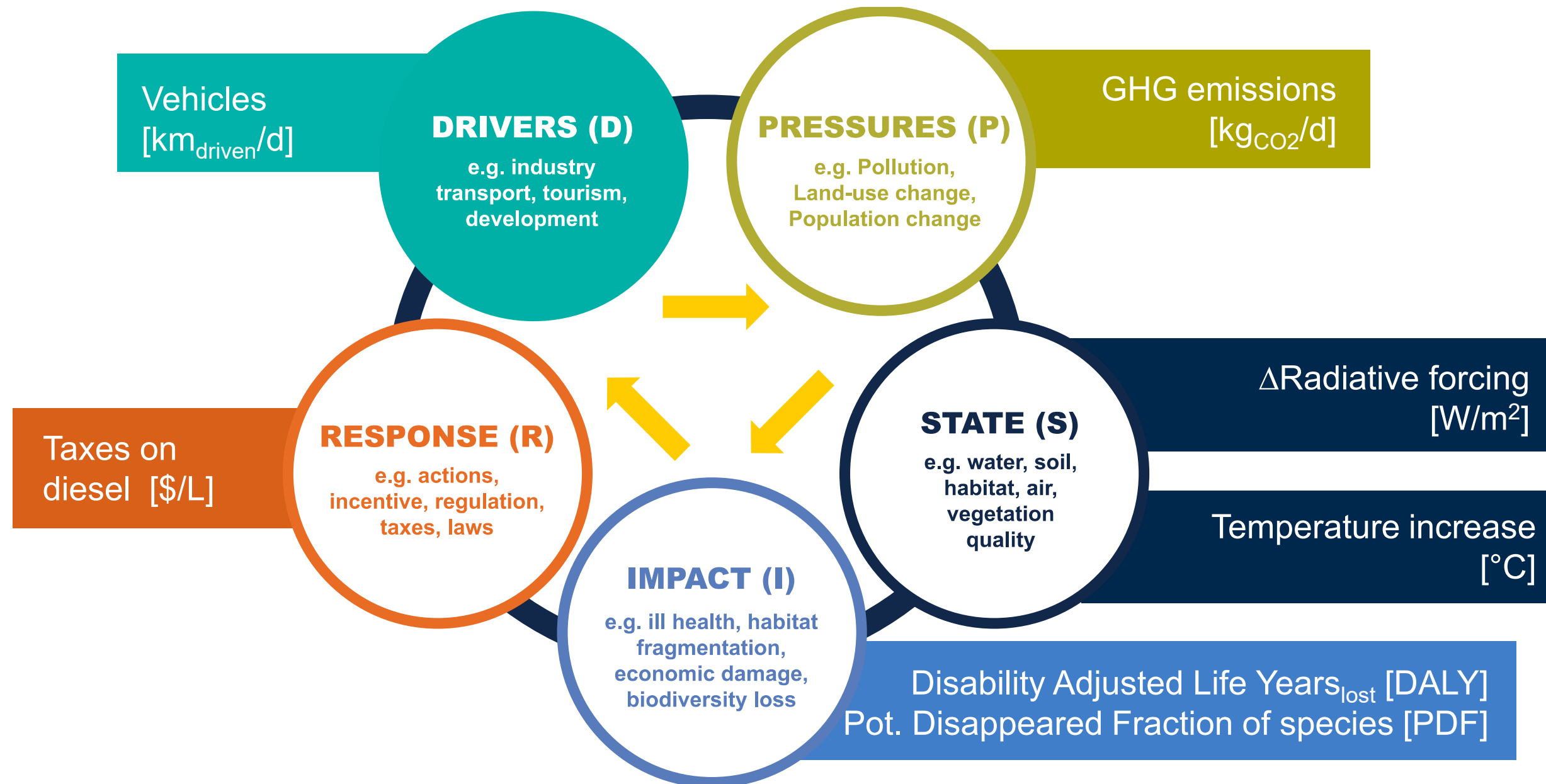
*consumption based emissions

Average DK carbon footprint per capita

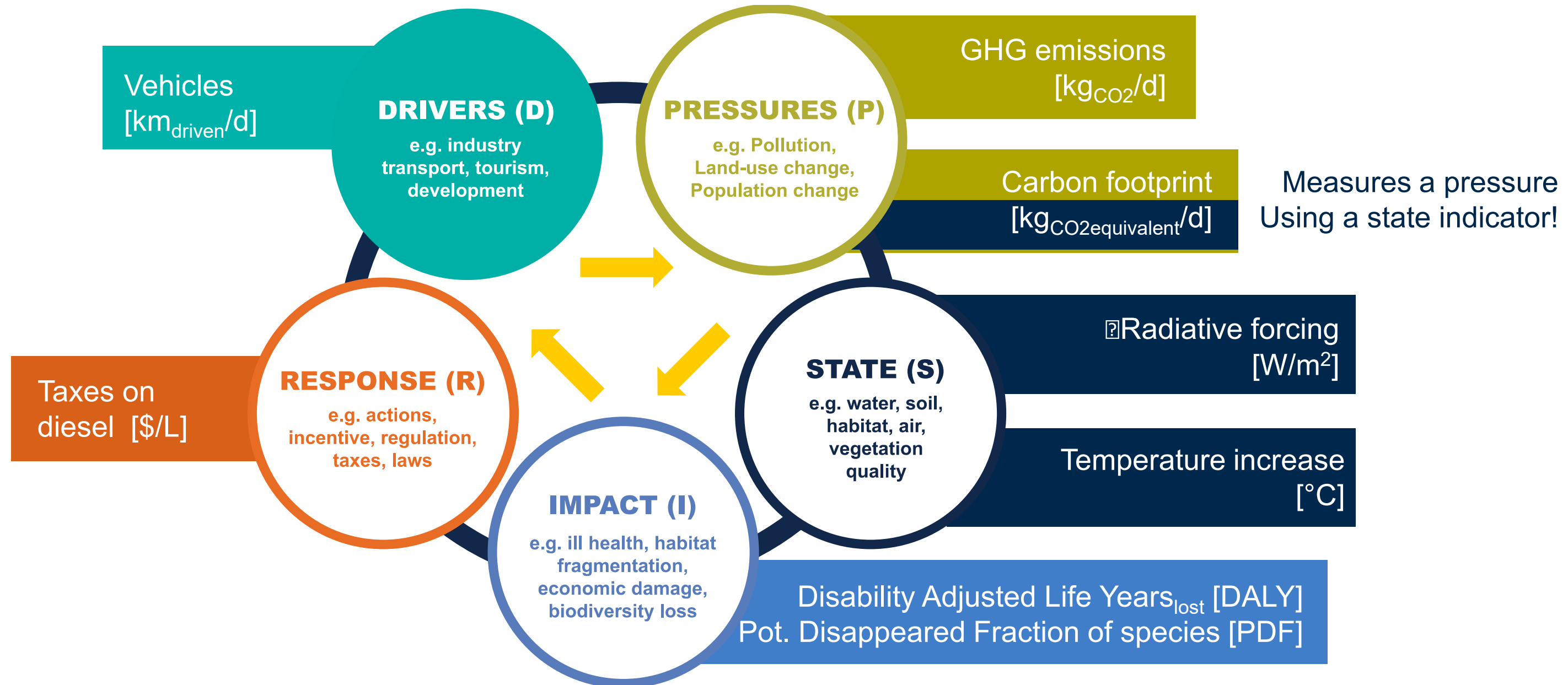
Substance		Unit kg _i	Emissions kg _i /pers/y	GWP100 kg _{CO2e} /kg _i	GW score kg _{CO2e} /pers/y
Carbon dioxide	fossil	kg CO ₂	8000*	1	8000
Methane	biogenic	kg CH ₄	46.1	27.0	1250
Methane	fossil	kg CH ₄	4.7	29.8	140
Nitrous Oxide*		kg N ₂ O	3.2	273	870
HFCs, PFCs,SF6					80
TOTAL kg _{CO2e} /pers/y					10340
TOTAL kg _{CO2e} /pers/d					28.3

*consumption based emissions

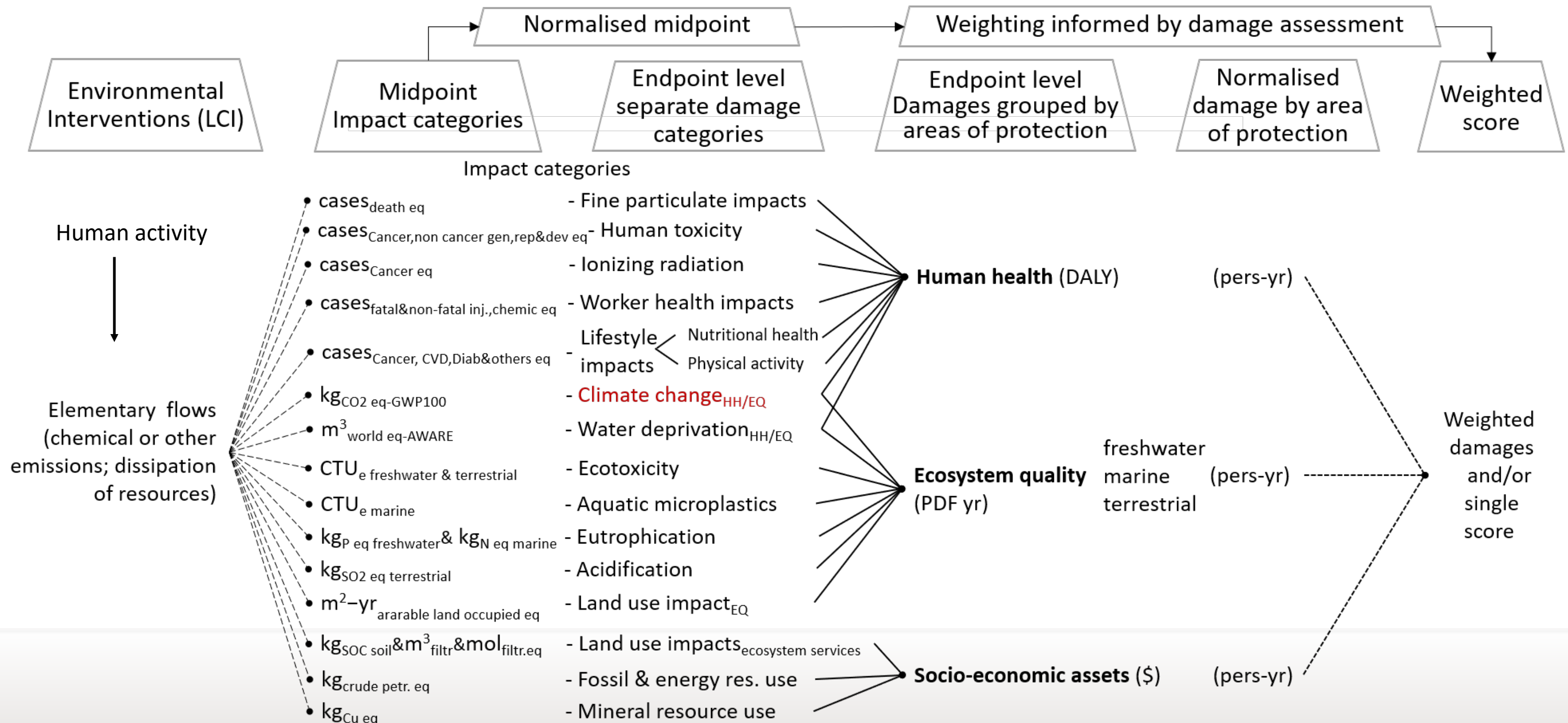
At which level of the cause effect chain is the carbon footprint?



At which level of the cause effect chain is the carbon footprint?



UNEP - GLAM: the global life cycle impact assessment framework





Determining the carbon footprint of human activity

Gasoline vs electric vehicles

Olivier Jolliet, DTU-Sustain, Quantitative Sustainability Assessment



Scenario comparison and interpretation

Interpretation Principles

Olivier Jolliet, DTU-Sustain, Quantitative Sustainability Assessment

Learning Objectives

- ▶ Define a process to establish a carbon footprint
- ▶ Introduce you to the 1210x QMAS Sustainability tool
- ▶ Compare CO₂ for various materials and processes
- ▶ Perform the carbon footprint of a gasoline versus an electric car
- ▶ Interpret results, comparing scenario performances and identifying hot spots

Case study: Gasoline vs electric vehicle



Building the carbon footprint

- 1 *Functional unit (FU):* Identify the functional basis for comparing the different scenarios. Footprints are calculated per functional unit (FU representative of **what service you want, not a ratio**)



What is a suitable functional unit to compare these vehicles?

Functional unit of a car

Functional unit:

What	How much	How well	Where	How long
“Transporting	1 passenger safely		in Europe	over 1 km commuting
1 pers-km				

Key parameter: Vehicle occupancy

On average: 1.1 passenger for commuting

→ Default scenario 1 passenger in all types of vehicle

→ 1 alternative scenario of electric car with 2 occupants

Building the carbon footprint

- 1 *Functional unit (FU)*: Identify the functional basis for comparing the different scenarios. Footprints are calculated per functional unit (FU representative of what service you want, not a ratio)
- 2 *Collect composition and consumption data*: collect the inputs to product manufacturing and its consumption during operation and calculate each input/FU (what you need to provide the service, in e.g. $\text{kg}_{\text{input}}/\text{FU}$, kWh/FU)



What are the main inputs data to collect for each vehicles?

The 1210x QMAS Sustainability tool – Carbon footprint

Carbon footprint tool

1

Functional Unit:

“Transporting 1 passenger safely in Europe for 1 km commuting”

FU

1 pers-km

Transformation multiplier

1000

t/kg, kg/g...

Conversion factor

3.6

MJ/kWh

Conversion efficiency - Electricity

25%

Conversion efficiency -Heat

80%

System 1: Gasoline car

2

Data

Lifespan of Gasoline car

150000

km

Weight of car

1422.1017

kg

Occupancy

1

pers/V

2

Material

Incineration mix - ICEV

42.5

MJ/kg

List of Processes	Process ID	Corresponding Eco-invent process	Location	Quantity per unit	Unit	Quantity per FU	Unit	Carbon footprint [kg CO2eq/Unit]	Carbon footprint [kg CO2eq/FU]	fraction carbon footprint
Manufacturing (Materials + Processing)				Per vehicle		Per FU				
steel production, low-alloyed, hot rolled, RER/Europe		#N/A	#N/A	1040	kg		#N/A	#N/A		#N/A
sheet rolling, steel	20338	sheet rolling, steel	RER	780	kg	5.20E-03	kg	0.32	1.64E-03	#N/A
aluminium production, primary, EU27 & EFTA	502	aluminium production, primary, ingot	IAI Arc	55	kg		kg	7.39	0.00E+00	#N/A
Copper & other metals (Zn, Ni, Pb)	20335	sheet rolling, copper	RER	38	kg	2.53E-04	kg	0.48	1.20E-04	#N/A
platinum	14151	platinum to generic market for metal catalyst for	GLO	1.70E-03	kg	1.13E-08	kg	45758.65	5.19E-04	#N/A
polyethylene, polypropylene & PVC	17333	market for polyethylene, high density, granulate	GLO	175	kg	1.17E-03	kg	2.83	3.31E-03	#N/A
synthetic rubber	21656	market for synthetic rubber	GLO	46	kg	3.07E-04	kg	3.17	9.73E-04	#N/A
glas	9020	market for flat glass, uncoated	RoW	32	kg	2.13E-04	kg	1.06	2.27E-04	#N/A
electronics, for control units	8391	electronics production, for control units	RER	6.1	kg	4.07E-05	kg	31.76	1.29E-03	#N/A
ethylene, paints, fluids	8639	market for ethylene	RER	30	kg	2.00E-04	kg	1.82	3.65E-04	#N/A
tap water	21711	tap water production, conventional treatment	RoW	3400	kg	2.27E-02	kg	0.00	9.42E-06	#N/A
heat, district or industrial, natural gas	10352	heat production, natural gas, at industrial furnace	Europe	2400	MJ	1.60E-02	MJ	0.08	1.21E-03	#N/A
electricity, medium voltage	7666	#N/A	#N/A	2250	kWh	1.50E-02	#N/A	#N/A	#N/A	#N/A
Transport				Per vehicle		Per FU				
transport, freight train	22465	market for transport, freight, train, fleet average	Europe	371	km	3.52E-03	metric ton*km	0.05	1.64E-04	#N/A
transport, freight, lorry 16-32 metric ton	22337	market for transport, freight, lorry, 16-32 metric	RER	39	km	3.70E-04	metric ton*km	0.19	7.01E-05	#N/A
transport, freight, sea, container ship	22445	market for transport, freight, sea, container ship	GLO	8000	km	7.58E-02	metric ton*km	0.01	7.78E-04	#N/A
Use phase				Per V-km		Per FU				
Supply chain petrol, low-sulfur	16715	market for petrol, low-sulfur	Europe	0.0619	kg		kg	1.04	0.00E+00	#N/A
Direct combustion	100001	Emissions & operation of vehicle	RER	0.0619	kg petrol		kg petrol	3.43	0.00E+00	#N/A
Disposal				Per vehicle		Per FU				
plastic mixture incineration	23993	treatment of waste plastic, mixture, municipal in	CH	221.00	kg	1.47E-03	kg	2.37	3.49E-03	#N/A
Glass to landfill	23563	treatment of waste glass, inert material landfill	CH	283.00	kg	1.89E-03	kg	0.01	1.05E-05	#N/A
Avoided electricity plastic incineration	7666	#N/A	#N/A	221.00	kWh	-4.35E-03	#N/A	#N/A	#N/A	#N/A
								Total kgCO2eq/FU	#N/A	

Sustainability Tool Instructions

This document is an instruction on how to make an absolute sustainability assessment using the "Sustainability tool". It is meant as a companion to the tool, and should be consulted in case of confusion, rather than read from start to finish. For some explanations, examples will be provided referring to an example case study comparing the absolute sustainability of light bulbs to that of candles. To use the 'Sustainability tool', you need a functional unit and a list of materials and processes that are included in the cradle-to-grave life cycle of the products or processes studied.

1 Ecoinvent Database

In the sheet 'Ecoinvent Data', you can find a copy of the Ecoinvent database, with about 25 000 processes. For each process, several environmental indicators are given, as seen in Figure 1.

carbon footprint	remaining human health damage	remaining ecosystem quality damage	fossil and nuclear energy use	water scarcity footprint	global warming potential (GWP100)	remaining human health damage
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How to perform a carbon footprint

Amount of inputs
per functional Unit
(e.g. kg/pers-km)

Q

$\times$

*GHG Emission
factors per
unit input:*

E

$=$

*Carbon footprint
per FU*

F

$\text{kg}_{\text{steel}}/\text{pers-km}$

$\times$

$\text{kg}_{\text{CO2eq}}/\text{kg}_{\text{steel}}$

$=$

$\text{kg}_{\text{CO2eq}}/\text{FU}$

$\text{kg}_{\text{gas}}/\text{pers-km}$

$\times$

$\text{kg}_{\text{CO2eq}}/\text{kg}_{\text{gas}}$

$=$

$\text{kg}_{\text{CO2eq}}/\text{FU}$

You determine the inputs per FU!

Database
& spreadsheet



You interpret the spreadsheet results

- Look at main contributions
- Use graph on the right
- Identify potential improvements

Building the carbon footprint

- 1** *Functional unit (FU):* Identify the functional basis for comparing the different scenarios. Footprints are calculated per functional unit (FU representative of what service you want, not a ratio)
- 2** *Collect composition and consumption data:* collect the inputs to product manufacturing and its consumption during operation and **calculate each input/FU** (what you need to provide the service, in e.g. $\text{kg}_{\text{input}}/\text{FU}$, kWh/FU)
- 3** *Find GHG emission factors per unit input:*
Find in databases for each input the GHG emission in $\text{kg}_{\text{CO}_2 \text{ equivalent}}$ per unit input
- 4** *Calculate energy balance and carbon footprint per FU:* Multiply 2 and 3 to obtain the contribution of each input to the energy and carbon footprint ($\text{MJ}_{\text{non renewable primary energy}}/\text{FU}$, $\text{kg}_{\text{CO}_2 \text{ equivalent}}/\text{FU}$)
- 5** *Total footprint and interpretation:* Sum up the footprint of each input to calculate the total energy and carbon footprint of the considered service or product

CO₂ balance gasoline: calculate each input/FU

Carbon footprint tool

1

Functional Unit:	"Transporting 1 passenger safely in Europe for 1 km commuting"
FU	1 pers-km

Transformation multiplier	1000	t/kg, kg/g...
Conversion factor	3.6	MJ/kWh
Conversion efficiency - Electricity	25%	
Conversion efficiency -Heat	80%	

System 1: Gasoline car

3

Data	Unit
Lifespan of Gasoline car	km
Weight of car	1422.1017 kg
Occupancy	1 pers/V

2

Inc. ICEV	42.5	MJ/kg
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List of Processes	Process ID	Corresponding Eco-invent process	Location	Quantity per unit	Unit	Quantity per FU	Unit	Carbon footprint [kg CO2eq/Unit]	Carbon footprint [kg CO2eq/FU]	fraction carbon footprint
Manufacturing (Materials + Processing)		Manufacturing (Materials + Processing)		Per vehicle		Per FU				
steel production, low-alloyed, hot rolled, RER		#N/A	#N/A	1040	kg		#N/A	#N/A		#N/A
sheet rolling, steel	20338	sheet rolling, steel	RER	780	kg	5.20E-03	kg	0.32	1.64E-03	#N/A
aluminium production, primary, ingot		#N/A	#N/A	55	kg		#N/A	#N/A		#N/A
Copper & other metals (Zn, Ni, Pb)	20335	sheet rolling, copper	RER	38	kg	2.53E-04	kg	0.48	1.20E-04	#N/A
platinum	14151	platinum to generic market for metal catalyst for ca	GLO	1.70E-03	kg	1.13E-08	kg	45758.65	5.19E-04	#N/A
polyethylene, polypropylene & PVC	17333	market for polyethylene, high density, granulate	GLO	175	kg	1.17E-03	kg	2.83	3.31E-03	#N/A
synthetic rubber	21656	market for synthetic rubber	GLO	46	kg	3.07E-04	kg	3.17	9.73E-04	#N/A
glas	9020	market for flat glass, uncoated	RoW	32	kg	2.13E-04	kg	1.06	2.27E-04	#N/A
electronics, for control units	8391	electronics production, for control units	RER	6.1	kg	4.07E-05	kg	31.76	1.29E-03	#N/A
ethylene, paints, fluids	8639	market for ethylene	RER	30	kg	2.00E-04	kg	1.82	3.65E-04	#N/A
tap water	21711	tap water production, conventional treatment	RoW	3400	kg	2.27E-02	kg	0.00	9.42E-06	#N/A
heat, district or industrial, natural gas	10352	heat production, natural gas, at industrial furnace	Europe	2400	MJ	1.60E-02	MJ	0.08	1.21E-03	#N/A
electricity, medium voltage	7666	#N/A	#N/A	2250	kWh	1.50E-02	#N/A	#N/A	#N/A	#N/A
Transport		Transport		Per vehicle		Per FU				
transport, freight train	22465	market for transport, freight, train, fleet average	Europe	371	km	3.52E-03	metric ton*km	0.05	1.64E-04	#N/A
transport, freight, lorry 16-32 metric ton	22337	market for transport, freight, lorry, 16-32 metric ton	RER	39	km	3.70E-04	metric ton*km	0.19	7.01E-05	#N/A
transport, freight, sea, container ship	22445	market for transport, freight, sea, container ship, h	GLO	8000	km	7.58E-02	metric ton*km	0.01	7.78E-04	#N/A
Use phase		Use phase		Per V-km		Per FU				
Supply chain petrol, low-sulfur		market for petrol, low-sulfur	Europe	0.0619	kg		kg	1.04	0.00E+00	#N/A
Direct combustion	100001	Emissions & operation of vehicle	RER	0.0619	kg petrol		kg petrol	3.43	0.00E+00	#N/A
Disposal		Disposal		Per vehicle		Per FU				
plastic mixture incineration	23993	treatment of waste plastic, mixture, municipal incin	CH	221.00	kg	1.47E-03	kg	2.37	3.49E-03	#N/A
Glass to landfill	23563	treatment of waste glass, inert material landfill	CH	283.00	kg	1.89E-03	kg	0.01	1.05E-05	#N/A
Avoided electricity plastic incineration	7666	#N/A	#N/A	221.00	kWh	-4.35E-03	#N/A	#N/A	#N/A	#N/A
								Total kgCO2eq/FU	#N/A	

The ecoinvent database 25000+ processes – in the ecoinvent tab

COPYRIGHTS: The present Ecoinvent data can only be used within DTU courses or theses. It cannot be distributed				Method	IMPACT World+ v2.1, footprint version
				Category	climate change
				Indicator	carbon footprint
Process #	Activity Name	Geography	Reference Product Name	Reference Product Amount	kg CO2-Eq
7658	market group for electricity, low voltage	ON-5000	electricity, low voltage	kWh	0.31
7659	market group for electricity, low voltage	Canada without Quebec	electricity, low voltage	kWh	0.29
7660	market group for electricity, low voltage	ENTSO-E	electricity, low voltage	kWh	0.32
7661	market group for electricity, low voltage	Europe without Switzerland	electricity, low voltage	kWh	0.34
7662	market group for electricity, low voltage	GLO	electricity, low voltage	kWh	0.69
7663	market group for electricity, low voltage	IN	electricity, low voltage	kWh	1.47
7664	market group for electricity, low voltage	RAF	electricity, low voltage	kWh	0.85
7665	market group for electricity, low voltage	RAS	electricity, low voltage	kWh	0.88
7666	market group for electricity, low voltage	RER	electricity, low voltage	kWh	0.33
7667	market group for electricity, low voltage	RLA	electricity, low voltage	kWh	0.35
7668	market group for electricity, low voltage	RME	electricity, low voltage	kWh	0.91
7669	market group for electricity, low voltage	RNA	electricity, low voltage	kWh	0.45
7670	market group for electricity, low voltage	UCTE	electricity, low voltage	kWh	0.36
7671	market group for electricity, low voltage	US	electricity, low voltage	kWh	0.49
7672	natural gas, burned in micro gas turbine, 100kWe	CH	electricity, low voltage	kWh	0.73
7673	natural gas, burned in polymer electrolyte membrane fuel cell	CH	electricity, low voltage	kWh	0.71
7674	natural gas, burned in polymer electrolyte membrane fuel cell	RoW	electricity, low voltage	kWh	0.69
7675	natural gas, burned in solid oxide fuel cell 125kWe, future	CH	electricity, low voltage	kWh	0.52
7676	natural gas, burned in solid oxide fuel cell 125kWe, future	RoW	electricity, low voltage	kWh	0.51
7677	natural gas, burned in solid oxide fuel cell, with micro gas turbine	CH	electricity, low voltage	kWh	0.44
7678	natural gas, burned in solid oxide fuel cell, with micro gas turbine	RoW	electricity, low voltage	kWh	0.43
7679	sawing and planing, paraná pine, kiln dried	BR	electricity, low voltage	kWh	0.33
7680	sawing and planing, paraná pine, kiln dried	RoW	electricity, low voltage	kWh	0.62
7681	wood pellets, burned in stirling heat and power co-generation unit	CH	electricity, low voltage	kWh	0.12
7682	wood pellets, burned in stirling heat and power co-generation unit	RoW	electricity, low voltage	kWh	0.14
7683	electricity production, photovoltaic, 3kWp slanted-roof installation	CH	electricity, low voltage, renewable	kWh	0.08
7684	electricity production, photovoltaic, 3kWp slanted-roof installation	CH	electricity, low voltage, renewable	kWh	0.09
7685	electricity voltage transformation from medium to low voltage, 100kV to 10kV	CH	electricity, low voltage, renewable	kWh	0.01
7686	market for electricity, low voltage, renewable energy products	CH	electricity, low voltage, renewable	kWh	0.02

Olivier Jean Joliet:
You might use the arrow on the bottom right of this field to then enter keywords and search for processes of

Regions:

RER, UCTE: Europe

ROW: Rest of the World

GLO: Global

CA: Canada

CN: China

IN: India

US: USA

Use “market” processes when available

Identify the # number

→ Copy in selected Processes & in Carbon footprint

CO₂ balance gasoline: calculate footprint/input

3

Carbon footprint tool										
Functional Unit		Transporting 1 passenger safely in Denmark for 1 km commuting				Transformation multiplier		1000	t/kg, kg/g...	
FU		1 pers-km				Conversion factor		3.6	MJ/kWh	
						Conversion efficiency - Electricity		25%		
						Conversion efficiency -Heat		80%		
System 1: Gasoline car										
Data		Unit		Material		Energy content		Unit		
Lifespan of Gasoline car		150000	km	Incineration mix - ICEV		42.5		MJ/kg		
Weight of car		1422.1017	kg							
Occupancy		1	pers/V							
List of Processes	Process ID	Corresponding Eco-invent process	Location	Quantity per unit	Unit	Quantity per FU	Unit	Carbon footprint [kg CO2eq/Unit]	Carbon footprint [kg CO2eq/FU]	fraction carbon footprint
Manufacturing (Materials + Processing)		Manufacturing (Materials + Processing)		Per vehicle		Per FU				
steel production, low-alloyed, hot rolled, RER	21143	steel production, low-alloyed, hot rolled	RER	1040	kg	6.93E-03	kg	2.27	1.57E-02	5.04%
sheet rolling, steel	20338	sheet rolling, steel	RER	780	kg	5.20E-03	kg	0.32	1.64E-03	0.53%
aluminium production, primary, ingot	502	aluminium production, primary, ingot	IAI Are	55	kg	3.67E-04	kg	7.39	2.71E-03	0.87%
Copper & other metals (Zn, Ni, Pb)	20335	sheet rolling, copper	RER	38	kg	2.53E-04	kg	0.48	1.20E-04	0.04%
platinum	14151	platinum to generic market for metal catalyst for catalytic	GLO	1.70E-03	kg	1.13E-08	kg	45758.65	5.19E-04	0.17%
polyethylene, polypropylene & PVC	17333	market for polyethylene, high density, granulate	GLO	175	kg	1.17E-03	kg	2.83	3.31E-03	1.06%
synthetic rubber	21656	market for synthetic rubber	GLO	46	kg	3.07E-04	kg	3.17	9.73E-04	0.31%
glas	9020	market for flat glass, uncoated	RoW	32	kg	2.13E-04	kg	1.06	2.27E-04	0.07%
electronics, for control units	8391	electronics production, for control units	RER	6.1	kg	4.07E-05	kg	31.76	1.29E-03	0.41%
ethylene, paints, fluids	8639	market for ethylene	RER	30	kg	2.00E-04	kg	1.82	3.65E-04	0.12%
tap water	21711	tap water production, conventional treatment	RoW	3400	kg	2.27E-02	kg	0.00	9.42E-06	0.00%
heat, district or industrial, natural gas	10352	heat production, natural gas, at industrial furnace >100kW	Europe	2400	MJ	1.60E-02	MJ	0.08	1.21E-03	0.39%
electricity, medium voltage	7666	market group for electricity, low voltage	RER	2250	kWh	1.50E-02	kWh	0.33	4.96E-03	1.59%
Transport		Transport		Per vehicle		Per FU				
transport, freight train	22465	market for transport, freight, train, fleet average	Europe	371	km	3.52E-03	metric ton*km	0.05	1.64E-04	0.00
transport, freight, lorry 16-32 metric ton	22337	market for transport, freight, lorry, 16-32 metric ton, diesel	RER	39	km	3.70E-04	metric ton*km	0.19	7.01E-05	0.00
transport, freight, sea, container ship	22445	market for transport, freight, sea, container ship, heavy fuel oil	GLO	8000	km	7.58E-02	metric ton*km	0.01	7.78E-04	0.00
Use phase		Use phase		Per V-km		Per FU				
Supply chain petrol, low-sulfur	16715	market for petrol, low-sulfur	Europe	0.0619	kg	6.19E-02	kg	1.04	6.42E-02	0.21
Direct combustion	100001	Emissions & operation of vehicle	RER	0.0619	kg petrol	6.19E-02	kg petrol	3.43	2.12E-01	0.68
Disposal		Disposal		Per vehicle		Per FU				
plastic mixture incineration	23993	treatment of waste plastic, mixture, municipal incineration	CH	221.00	kg	1.47E-03	kg	2.37	3.49E-03	0.01
Glass to landfill	23563	treatment of waste glass, inert material landfill	CH	283.00	kg	1.89E-03	kg	0.01	1.05E-05	0.00
Avoided electricity plastic incineration	7666	market group for electricity, low voltage	RER	221.00	kWh	-4.35E-03	kWh	0.33	-1.44E-03	0.00
								Total kgCO2eq/FU	0.312	

The Ecoinvent database



The broadest Unit Process level database worldwide

- Energy systems - non renewable & renewable
- Materials and building
- Chemicals and metals
- Transports
- Packaging
- Waste treatment
- Agriculture
- Electronics

100 Inventory flows for 20000 unit processes

Non-renewable primary energy and CO₂ aggregated inventory flows

	CO ₂ (kg per unit)
Energy Carriers	1 kWh electricity (Europe)
	0.39
	1 kWh electricity (USA)
	0.71
	1 kWh electricity (Japan)
	0.53
	1 kWh electricity (Switzerland)
	0.11
Energy Carriers	1 kWh electricity (China)
	0.98
	1L gasoline (no combustion)
	0.49
	1L gasoline (with combustion)
	2.88
	1 kg light oil (42.7 MJ final)
	3.71
Transport	1,000 km kg transport 16-32 ton Lorry
	0.19
	1 pers • km by train (Intercity)
	0.06
Transport	1 pers • km by plane (Intercontinental)
	0.19
Transport	1 pers • km by car
	0.17

Non-renewable primary energy and CO₂ Aggregated inventory flows

		CO ₂ (kg per unit)
Materials	1 m ³ concrete	257
	1 m ³ water	0.44
	1 kg steel, low alloy	1.63
	1 kg copper	1.86
	1 kg primary aluminium	7.21
	1 kg recycled aluminium	1.32
	1 kg platinum	70,000
	1 kg paper newsprint	1.22
	1 kg glass	1.06
	1 kg polyethylene HDPEs	1.56
Material processing	1 kg plastic film extrusion	0.50
	1 kg injection moulding	1.24
End of life	1 kg landfilled steel	0.00657
	1 kg landfilled aluminium	0.02010
	1 kg incinerated municipal solid waste	0.50
	1 kg incinerated polyethylene	2.99

The ecoinvent database 25000+ processes – in the ecoinvent tab

COPYRIGHTS: The present Ecoinvent data can only be used within DTU courses or theses. It cannot be distributed				Method	IMPACT World+ v2.1, footprint version
				Category	climate change
				Indicator	carbon footprint
Process #	Activity Name	Geography	Reference Product Name	Reference Product Amount	kg CO2-Eq
7658	market group for electricity, low voltage	ON-5000	electricity, low voltage	kWh	0.31
7659	market group for electricity, low voltage	Canada without Quebec	electricity, low voltage	kWh	0.29
7660	market group for electricity, low voltage	ENTSO-E	electricity, low voltage	kWh	0.32
7661	market group for electricity, low voltage	Europe without Switzerland	electricity, low voltage	kWh	0.34
7662	market group for electricity, low voltage	GLO	electricity, low voltage	kWh	0.69
7663	market group for electricity, low voltage	IN	electricity, low voltage	kWh	1.47
7664	market group for electricity, low voltage	RAF	electricity, low voltage	kWh	0.85
7665	market group for electricity, low voltage	RAS	electricity, low voltage	kWh	0.88
7666	market group for electricity, low voltage	RER	electricity, low voltage	kWh	0.33
7667	market group for electricity, low voltage	RLA	electricity, low voltage	kWh	0.35
7668	market group for electricity, low voltage	RME	electricity, low voltage	kWh	0.91
7669	market group for electricity, low voltage	RNA	electricity, low voltage	kWh	0.45
7670	market group for electricity, low voltage	UCTE	electricity, low voltage	kWh	0.36
7671	market group for electricity, low voltage	US	electricity, low voltage	kWh	0.49
7672	natural gas, burned in micro gas turbine, 100kWe	CH	electricity, low voltage	kWh	0.73
7673	natural gas, burned in polymer electrolyte membrane fuel cell	CH	electricity, low voltage	kWh	0.71
7674	natural gas, burned in polymer electrolyte membrane fuel cell	RoW	electricity, low voltage	kWh	0.69
7675	natural gas, burned in solid oxide fuel cell 125kWe, future	CH	electricity, low voltage	kWh	0.52
7676	natural gas, burned in solid oxide fuel cell 125kWe, future	RoW	electricity, low voltage	kWh	0.51
7677	natural gas, burned in solid oxide fuel cell, with micro gas turbine	CH	electricity, low voltage	kWh	0.44
7678	natural gas, burned in solid oxide fuel cell, with micro gas turbine	RoW	electricity, low voltage	kWh	0.43
7679	sawing and planing, paraná pine, kiln dried	BR	electricity, low voltage	kWh	0.33
7680	sawing and planing, paraná pine, kiln dried	RoW	electricity, low voltage	kWh	0.62
7681	wood pellets, burned in stirling heat and power co-generation unit	CH	electricity, low voltage	kWh	0.12
7682	wood pellets, burned in stirling heat and power co-generation unit	RoW	electricity, low voltage	kWh	0.14
7683	electricity production, photovoltaic, 3kWp slanted-roof installation	CH	electricity, low voltage, renewable	kWh	0.08
7684	electricity production, photovoltaic, 3kWp slanted-roof installation	CH	electricity, low voltage, renewable	kWh	0.09
7685	electricity voltage transformation from medium to low voltage, 100kV to 10kV	CH	electricity, low voltage, renewable	kWh	0.01
7686	market for electricity, low voltage, renewable energy products	CH	electricity, low voltage, renewable	kWh	0.02

Olivier Jean Joliet:
You might use the arrow on
the bottom right of this field
to then enter keywords and
search for processes of

Regions:

RER, UCTE: Europe

ROW: Rest of the World

GLO: Global

CA: Canada

CN: China

IN: India

US: USA

Use “market” processes
when available

Identify the
number

→ Copy in
selected Processes &
in Carbon footprint

Building the energy balance and carbon footprint

- 1** *Functional unit (FU):* Identify the functional basis for comparing the different scenarios. Footprints are calculated per functional unit (FU representative of **what service you want, not a ratio**)
- 2** *Collect composition and consumption data:* collect the inputs to product manufacturing and its consumption during operation and calculate each input/FU (what you need to provide the service, in e.g. $\text{kg}_{\text{input}}/\text{FU}$, kWh/FU)
- 3** *Find resource consumption and GHG emission factors per unit input:* Find in databases for each input the $\text{MJ}_{\text{non renewable primary energy}}$ and GHG emission in $\text{kg}_{\text{CO2 equivalent}}$ per unit input
- 4** *Calculate energy balance and carbon footprint per FU:* Multiply 2 and 3 to obtain the contribution of each input to the energy and carbon footprint ($\text{MJ}_{\text{non renewable primary energy}}/\text{FU}$, $\text{kg}_{\text{CO2 equivalent}}/\text{FU}$)
- 5** *Total footprint and interpretation:* Sum up the footprint of each input to calculate the total energy and carbon footprint of the considered service or product

Interpretation question for each scenario

Which are the largest contribution to the environmental impact in terms of production phase, in terms of pollutant?

- The grouping of processes can be arbitrary and depends on how the system is modeled.
- Also consider large sets of individual processes that each appear to have small impacts but lead to substantial impacts when summed

Which life cycle stage has the highest potential to reduce impacts (also depending on feasibility)

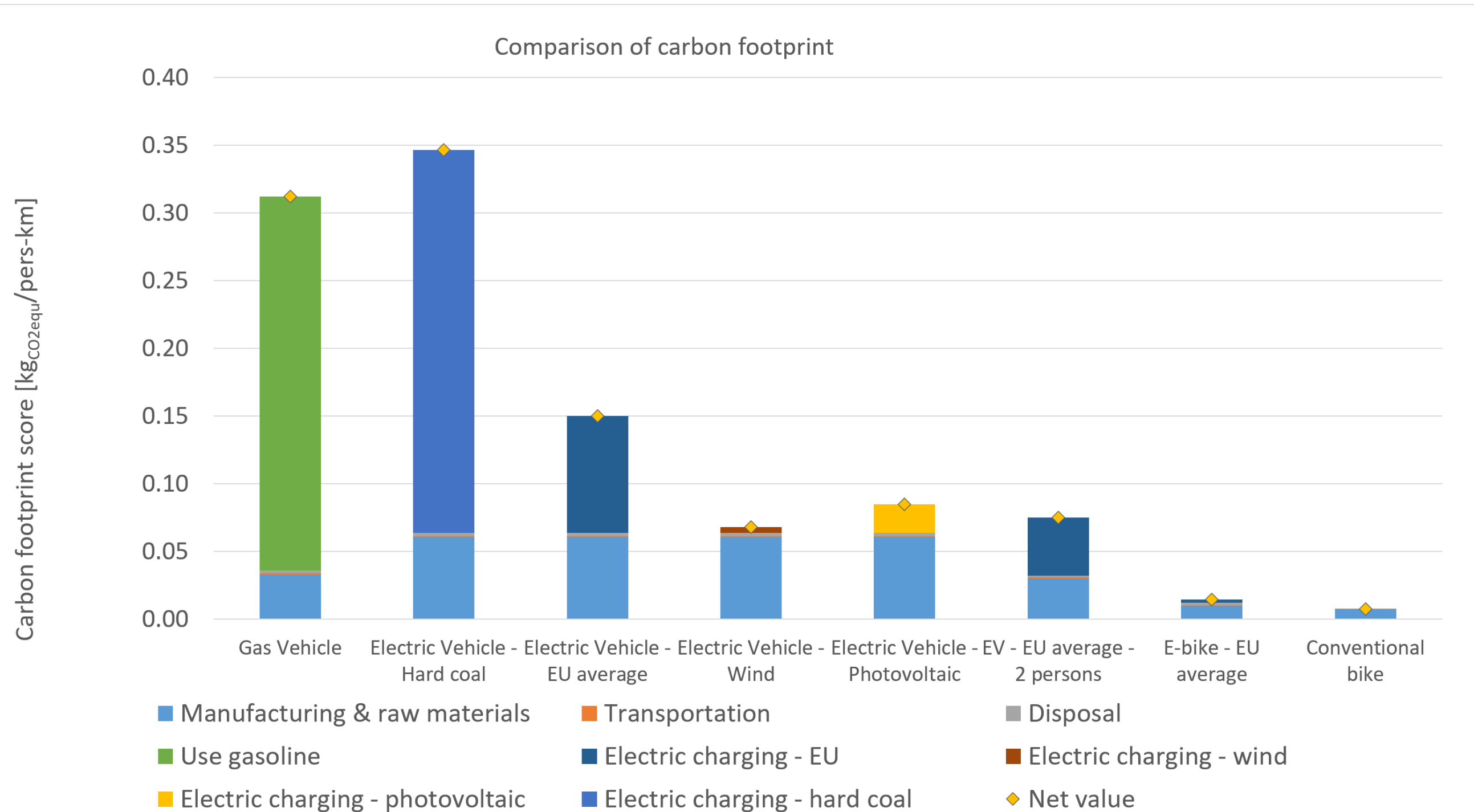
- In some win-win cases, both impacts and costs can be reduced at the same time.
- Even a limited low-cost intervention can be extremely efficient in reducing impacts.

- **If some results are surprising:** WHY? What are the underlying parameter components and stages responsible for it?

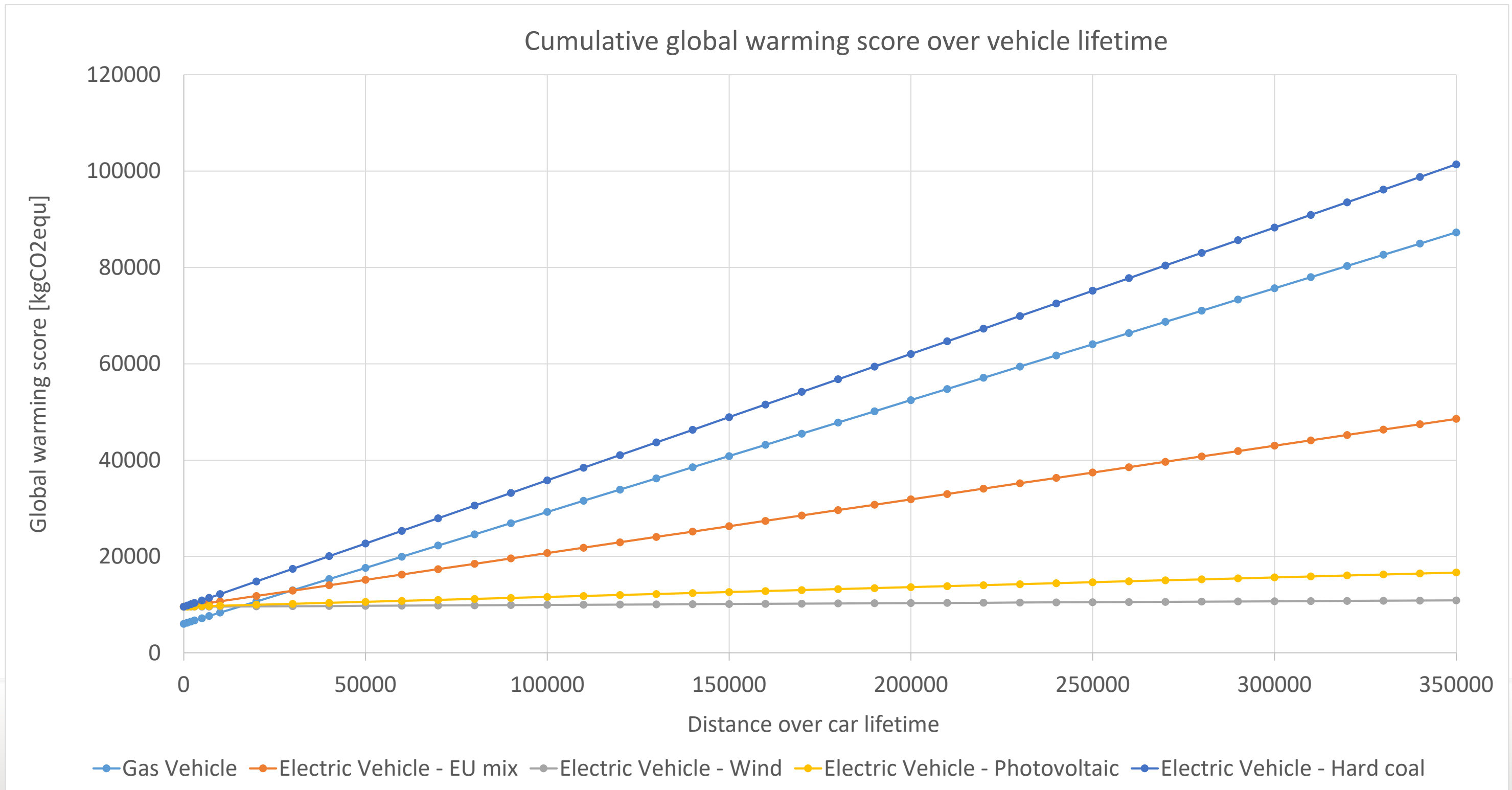
*Never let it go until you understand the rational for a difference or a surprising results.
Either you made a mistake ... or you will learn something!*



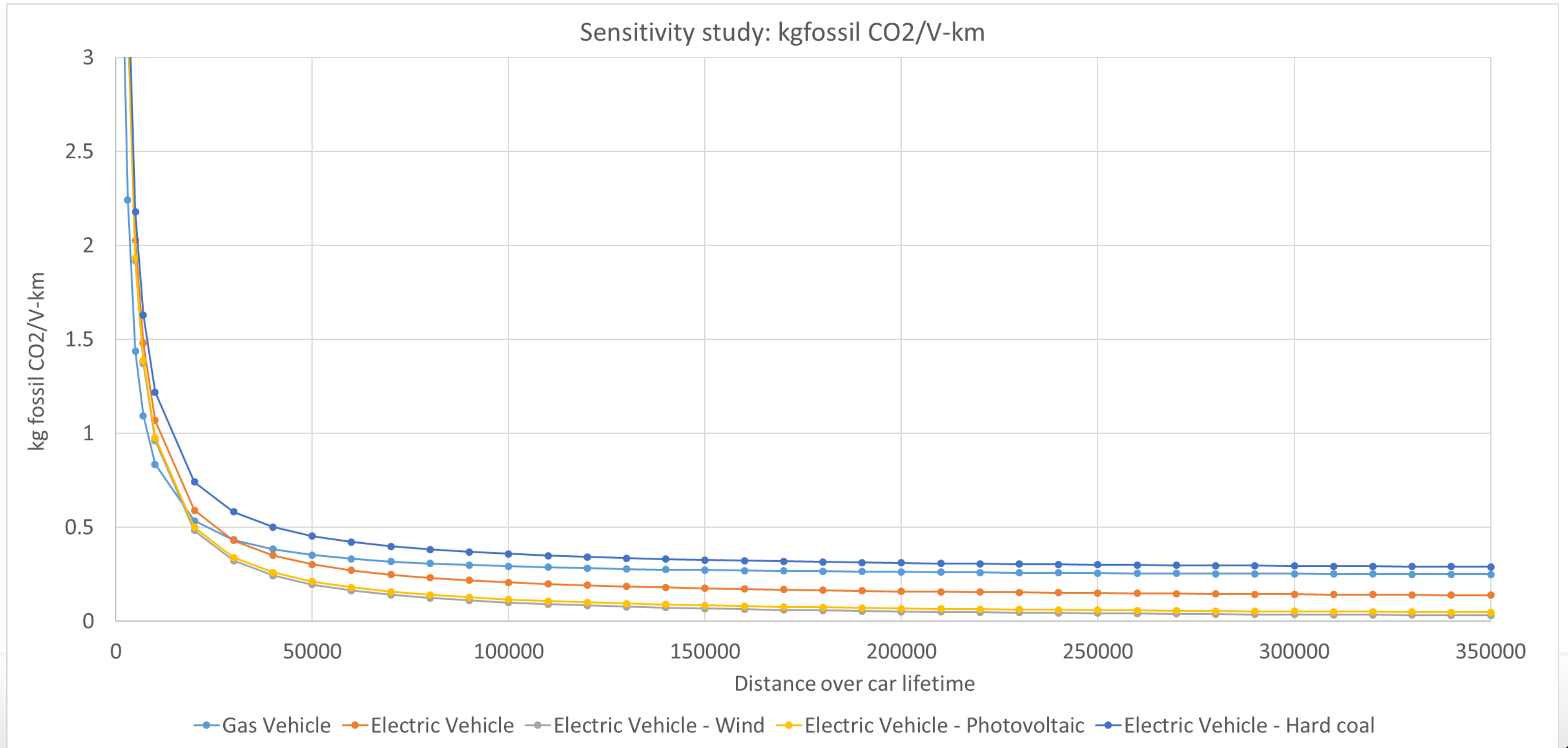
Carbon footprint: gasoline vs electric vehicle



Sensitivity study: cumulative carbon footprint as a function of vehicle lifetime



Sensitivity study: carbon footprint per FU as a function of vehicle lifetime



Comparison of normalized damage Impact World+ Life cycle impact Assessment

